

Part I. Shorter problems (9 points each). Give brief but to the point justification.

1. Let A and B be two 5×5 matrices over $\mathbb{Q}$ that both have the characteristic polynomial $x^5 + 3x + 3$. Prove that A and B are similar.
2. Compute the greatest common divisor of $5 + i$ and 13 in the ring $\mathbb{Z}[i]$ of Gaussian numbers.
3. Let R be a ring, and let M be a Noetherian left R -module. Let $f : M \rightarrow M$ be a surjective homomorphism. Prove that f is an isomorphism.
4. Prove that if I and J are ideals in $\mathbb{C}[x_1, \dots, x_n]$ such that IJ is a radical ideal, then $IJ = I \cap J$.
5. Let k be a field, and let R be a simple non-commutative k -algebra such that $\dim_k R = 4$. Prove that the center of R is k .
6. Determine the automorphism group of $C_2 \oplus C_4$.

II. Longer problems (12 points each). Do any *four* of the following problems (only 4 problems with highest scores will be counted).

1. Let R be the ring of upper-triangular $n \times n$ -matrices over $\mathbb{Z}/9\mathbb{Z}$. What is the Jacobson radical of R ? Classify irreducible R -modules.
2. a) For an element g of a finite group there is a unique decomposition $g = xy = yx$ where the order of x is a power of 2 and the order of y is odd.
(b) The map $g \mapsto x$ (where x is as in part (a)) is an endomorphism θ of forgetful functor **FiniteGroups** $\rightarrow$ **Sets**.
(c) The endomorphism θ does not lift to the category **Groups**: there is no endomorphism ψ of the forgetful functor **Groups** $\rightarrow$ **Sets** such that ψ restricted to **FiniteGroups** is θ .
3. Let V be an irreducible complex representation of S_n and let $V' = V \otimes \text{sign}$, where sign is the sign representation of S_n . Prove that the following are equivalent:
(i) V' is equivalent to V as S_n -representation;
(ii) The restriction of V to A_n is reducible;
(iii) $\chi_V(\sigma) = 0$ for any odd permutation σ , where χ_V is the character of V .
4. Prove that any group of order $255 = 3 \cdot 5 \cdot 17$ is cyclic.
5. Find all intermediate subfields F between $\mathbb{Q}$ and $K = \mathbb{Q}(\zeta_7)$, where ζ_7 is a primitive 7th root of unity. Express each such subfield as $F = \mathbb{Q}(\alpha)$ for some $\alpha \in K$.